DISCUSSION

SESSION 5A

O. D. KAZACHKOVSKII, *Session Chairman*

THE CHAIRMAN: I am honoured to be able to preside at this session, which is devoted to physical problems. To some extent it is a continuation of the session which took place yesterday. However, today attention will be paid not only to the fundamental problems but to the more practical aspects of fast reactor physics.

Presentation of Papers

F. STORRER—*Paper* 5A/1

I should like to say a few words about the experimental physics facilities at Cadarache.

The first one, which is a source reactor, Harmony, became critical for the first time in August 1965, and is now in operation. The critical facility, Mazurka, should be critical at the end of this year (1966), and the first loading will be a plutonium loading.

In the paper we have stressed the theoretical aspects of physics, which does not mean that we ignore the importance of fundamental data. For instance, in Mazurka we shall not be able to perform clean engineering mockups because there is not enough plutonium to simulate a large core, and also because, as a result of the flexibility of such machines, a perfect mockup is not possible—the lattice structure cannot be represented, for example. So we have to make more fundamental experiments, which are only good if they can be interpreted.

D. C. G. SMITH—*Paper* 5A/2

Several papers at this conference have stressed the uncertainties which attend the calculations which have to be made in support of a fast reactor design. But for some time we have judged it possible to estimate the important data with an accuracy which is adequate for reactor design purposes provided that the designers are aware of the margins and able to build in flexibility to accommodate the errors.

The paper deals with the particular methods which have been used to evaluate the data which have been required for the prototype fast reactor.

J. HIROTA—*Paper* 5A/3

Studies on fast reactor physics have recently been making progress in Japan. The paper mainly deals with theoretical and experimental work.

For the calculations of a neutron spectrum and effective cross-sections, the code ESELEM was developed. In this type of code, the fission spectrum and some cross-sections like inelastic cross-sections are usually approximated as constants in each coarse group. These approximations cause deep dips in the calculated spectrum. The dips make smaller the elastic and inelastic removal cross-sections, and then introduce an error in the calculation of reactivity effects. These drawbacks are eliminated in ESELEM.

In the study of the nuclear characteristics of fast reactors, zoned systems were examined. The reactivity changes due to sodium voids were evaluated for both the central and the annular region. The spectra were calculated by ESELEM.

To study the kinetics of fast reactors, the code TERESCOPE was developed to analyse reactor behaviour before the boiling of sodium. This code is written for the IBM-7044. By using this code, it is possible to get precise knowledge during the pre-accident stage of fast reactors.

Methods for measuring the various power coefficients of reactivity have been investigated and pulsed neutron experiments for non-multiplying and multiplying systems have been carried out to investigate the energy dependent time behaviour of fast neutrons in fast systems.

K. H. KÜSTERS—*Paper* 5A/4

The paper describes the fundamental procedures used at Karlsruhe to calculate fast reactors. (In his presentation, Dr Küsters described and illustrated the main point of the paper.)

P. B. F. EVANS—*Paper* 5A/5

This paper, which deals with the hardware for the prototype fast reactor, aims to do three things: first, to outline the principles of the control and instrumentation systems in the prototype fast reactor; second, to stress the problem areas; and, third, to indicate the development work aimed at providing solutions to those problems. The paper is in four sections discussing the control system, the core monitoring instrumentation, the sodium instrumentation and data handling and presentation.

Discussion

J. L. ROWLANDS (UKAEA): I should like to make some comments about the Pu^{239} Doppler coefficient calculations.

Papers 4A/4 and 5A/4 say that the Pu^{239} Doppler coefficient is not affected by U^{238} resonances. I think that there is now some evidence that for the resonance structure in Pu^{239} which was accepted until a few years ago the effect of overlap of U^{238} resonances on the effective cross-sections of Pu^{239} was a small one.

If one has a constant cross-section, then resonances in another material have no effect on the effective cross-section. I find it a little surprising, therefore, that Dr. Storrer (Paper 5A/1) finds the overlap effect to be important with broad Pu^{239} resonances. I wonder whether he has any physical explanation to offer for this effect.

I think that there are different definitions for the Pu^{239} component of the Doppler coefficient, and I should like to suggest that physicists try to adopt a single definition. "The change of reactivity with Pu^{239} temperature, keeping the temperatures of other constituents constant."

Dr. Storrer ignores the effect of the change of resonance shielding in uranium and plutonium isotopes with sodium density. We find this to be an effect which is larger than capture in sodium, and is larger than the heterogeneity effect which he observes, and in general, in the calculation of perturbation effects, we consider it important to take account of the changes in group constants caused by the perturbation to the within group weighting spectrum and, in particular, to resonance shielding factors.

A comment on Paper 5A/4. We have given some consideration to the Debye temperature, and believe that it is important to distinguish between the temperature associated with the oscillations of the uranium and plutonium atoms and the oscillations associated with the crystal structure as a whole. This latter Debye temperature is the one which enters into specific heat considerations, but the Debye temperature associated with the oscillation of the uranium and plutonium atoms is usually smaller—for example, 200–300°K.

H. H. HUMMEL (ANL): With regard to the Pu^{233} Doppler effect, it seems to me that perhaps Dr. Storrer's result is a consequence of this particular choice of Pu^{239} resonance parameters. He took the actual U^{238} resonance parameters, and for Pu^{239} he repeated what he found in the unresolved region as regard level spacings and widths. This represents a particular choice of parameters. He might have got different results had he made some different choice. I would ask what the rationale of this particular choice of parameters was?

I am not prepared to comment on the Debye temperatures.

With regard to the definition of the Pu^{239} component, I cannot go along with Mr. Rowlands on that, because the most important problem is that of the heterogeneous fuel element, and so the definition of where you hold the temperature is not applicable.

THE CHAIRMAN: It is a runaway accident where the separation is important.

H. H. HUMMEL: Yes, that is true. If you have fuel segregation of Pu^{239} and U^{238} within a fuel element, obviously this can be of importance. I am prepared to accept the definition that you work with effective cross-sections, including the flux correction factor, rather than the reaction rates that we have used in the past. I am sure that that is more correct, although I do not feel that it makes much difference to the final answer. However, it helps to avoid confusion. The main point to which I am addressing myself now is the choice by Dr. Storrer of the Pu^{239} parameters.

S. YIFTAH (Israel): Dr. Rowlands mentioned that I was hesitant about the effect of the resonances of Pu^{239} and U^{238}. I should like to clarify that. I mentioned in my paper (4A/2) that we used a Monte Carlo technique to study the effect of the interaction of the resonances between the fertile and the fissile isotopes, and the conclusion of this study indicates that the effect seems to be negligible. The details of this investigation are published* and in the framework of those calculations I am not hesitant in saying that the effect seems to be negligible.

K. H. KÜSTERS—*Paper* 5A/4: Dr. Storrer said that there is no great effect of heterogeneity for the total Doppler effect. That is true for a sodium reactor. However, if you have a steam cooled fast reactor with the more pronounced thermal tail, this is not true. In that case you should expect a greater effect on the Doppler coefficient.

F. STORRER—*Paper* 5A/1: I should like to say a few words in reply to Mr. Rowlands about how the overlap effect was calculated. This will also partly answer the question by Dr. Hummel.

We made the calculation at 1 keV for the U^{238} sample that we took. Actually, we took five different measured samples at neighbouring energies, and we shifted them in succession to 1 keV. So let us say that we had five different samples of uranium resonances. For the plutonium we took all the measured resonances by Uttley and by the French group at Saclay.

Instead of deriving statistical laws from these measured resonances and then refabricating an artificial sample, we took the measured sample directly. I do not see why this should give poor results, because we took all the measured resonance parameters and just shifted them up to 1 keV and there should not

* *Nucl. Sci. and Engng.* April 1966.

be much difference between 200 eV, the lower energy, and 1 keV. So I do not think we lost any information there.

To make sure that we did not have any artificial coincidences of resonances, we made the measurement with the five different U^{238} samples taken from neighbouring energies. So we have some statistics to see whether there are coincidences, and in the paper you will have the values for each sample of U^{238} resonances. So we took actual measured quantities without going through statistical laws.

We used the narrow resonance approximation, which I think is good. I think that you say in your paper (4*A*/5) that it is good to within 10%. The calculation was performed numerically on a computer, and it took a long time.

We are quite confident that the method is good because, for instance, we performed a calculation for infinite dilution, and we found that the effective cross-sections were indeed temperature independent, as they should be. We also used a special technique for the boundary of the sample to make sure that we did not have boundary effects. We took a mirror resonance to take that into account. So I think that the method is good.

With regard to the question, does it make sense, we said that this reduction of the positive coefficient came essentially from the wide resonances. We say this because we have made a calculation, I think that there were eight measured resonances, of which five were wide, width almost 1 eV. We made a calculation separately with the seventy-five narrow resonances and with the five wide resonances, and we found that the negative effect came essentially from the wide resonances numerically. Why is this? First, this is not an overlap effect. The wide resonance is not Doppler broadened as the other ones are, and so the increase in the reaction rate is reduced because of this.

It is true that if you go to a very wide resonance so that there is no resonance any more, there can be no overlap effect. The effective cross-section is the cross-section. For very narrow ones there might not be any overlap. I think that the term "overlap" itself means they overlap, and the wider the resonances the more they overlap. Of course, if they are too wide there is no more Doppler effect. This may be a poor answer, but you asked for a physical argument. There is probably a width of the resonance for which the overlap effect is maximum. If you want to overlap, you must have some width, or the resonances do not overlap.

As to Dr. Hummel's remark about the definition of the Pu^{239} Doppler effect, I am not sure that one has to take a unique definition. Your definition (Paper 4A/4) is adequate for reactivity measurements or for excursions in the reactor. On the other hand, one might make a measurement of a change in the reaction rate, and then the adequate definition would be a different one—the change in the reaction rate in the plutonium due to the heating of the surrounding material.

It was pointed out that in our calculation of the sodium coefficient in the heterogeneous case we did not include the fact that the narrow resonance self-shielding was modified when the sodium was lost. However, what I wanted to do was essentially to present a method. We express in the heterogeneous situation the reactivity change as a function of changes in cross-section and changes of collision probabilities. So you can do anything with this formula—Doppler, sodium, and a mixture of the two. I just chose to present two sample calculations, and I did not pretend that these two effects are independent—that we can treat separately the sodium problems and the Doppler resonance problems. However, I think that the method can be used basically for any combination of perturbations.

K. H. KÜSTERS: I should like to ask Dr. Storrer whether he can treat the heterogeneity effect in respect of small zones, e.g. fuel plates, as well as he can treat the heterogeneity effects in larger zones in his core. Did he extend his perturbation methods to total sodium loss? If so, to what order was the perturbation extended, and what comparisons are available with his successive k-calculations?

F. STORRER: Are you asking me if the method was applicable to calculating reactivity effects of the lattice structure?

K. H. KÜSTERS: The microstructure rather than the macrostructure of zoned cores. Say that we have a half flooded reactor and this heterogeneity.

F. STORRER: No. This is definitely a matter for the first case, which is the first kind of problem. It is really an infinite medium or a fundamental mode calculation, so that we can make calculations for the two zones separately. In the transition case we cannot use the method as such.

With regard to the total sodium loss, the calculation that we performed was a first order perturbation calculation. The method can be carried to higher orders because the fundamental assumption of this method—it is the only essential assumption—is that of flat flux in the individual regions, and this assumption is certainly more valid in the sodium-out case than in the sodium-in case. So we can carry this perturbation method to an infinitely higher order and get more and more precision. We have not done it yet. We have done this for the first application of the method, not the temperature dependent effects but just the effect on reactivity of the heterogeneous structure. We have not yet done it for the Doppler and sodium effects, but it can be done.

But before we do we think that we should develop a method which would do it with narrower groups—with the ELMOE kind of groups. It would be useless to do it with the usual twenty or twenty-five coarse groups. We have to take into account the change of the within-group spectrum when you remove the sodium. Obviously, when there is no more sodium you do not have any more flux depression at 3 keV. The first step then is to extend this method to a few hundred groups, which I think is possible. There may be a problem in regard to the memory of the machine. I think that the method we used does just about the same as the British method does when many thousands of groups are used. We use the same formalism of collision probability, but we use a perturbation formalism. So our method is more analytic and may cause memory problems if we use many groups, because it is an analytic method. We work with matrices and it takes a great deal of storage. You work with an iterative procedure which reduces the storage space.

K. H. KÜSTERS: In reply to Mr. Rowland's remarks about the influence of the Debye temperature, the calculations which have been done are only for UO_2. The measurements of Dolling are for UO_2. We would just like to see what the effect is. I agree that segregation is very important.

N. V. KRASNOYAROV (Physico-Energetics Institute, Obninsk): We calculated the reactivity due to sodium loss by two methods—perturbation theory and direct calculation. We had a change of concentration of substances in the core and obtained a new value of the reactivity. The result by the direct method was twice that given by the perturbation theory method. However, in these two cores the Doppler effect also turned out to be different. This means that it is not sufficient to change the concentration of sodium only. It is necessary to change the constants of all the other elements. I refer here to their resonance shielding. It means that in the perturbation theory we should take into account changes of constants as well as the flux change and, connected with it, the change of constants of all the other elements. It is a complex problem to calculate these effects with the help of perturbation theory or even by direct methods, because it requires a very large change of constants. If you have a large number of groups in which you take into account different resonances, it will be easier to do this with the help of computers. But if the number of groups is not large, it will require a very complex calculation of group constants.

J. A. SCOTT (English Electric): Confidence has been expressed in the ability to predict reactivity and enrichment for reactors. This confidence is based on a comparison of measurements which are on cold clean assemblies in which the isotopic composition is well known, with the calculations based on data sets.

However, the designer is interested in knowing what is going to happen in a reactor which is going to burn up to a high level, say, a 10% burnup, with, as a consequence, a significant change in the isotopic composition of the fissile material, plus the production of fission products. There has been no mention of this in Papers 5A/2 and 5A/4. Do the authors feel confident in their ability to predict the isotopic changes that will take place during the high burnups, because it is not something which will be easily discovered by experiment? With a given fuel cycle perhaps one might find equally significant differences, if not larger differences.

D. C. G. SMITH—*Paper* 5A/2: Your first point is whether we can extrapolate to conditions of higher temperatures and higher operating conditions conditions of higher burnup. Your second question points to why we want to build a prototype fast reactor. It is difficult to determine the capture cross-section for higher isotopes. The best way is to build a reactor and see what the reactivity changes will be. Our calculations are uncertain, but, such as they are, they seem to indicate that we shall have ample control and that we shall be able to keep the reactor going until we get to these higher burnups.

The comparisons made last year also bear on this question. If you look at some of those comparisons, particularly the very dilute core calculations, they seem to show very large differences in the critical mass as calculated by differing groups. But with very large reactors, a large change in critical mass is reflected as a very small change in enrichment. Therefore, here again we do not expect our enrichments to be very far wrong even though they have been calculated by different people.

Lastly, we think that we have sufficient margin in the design, partly by interchange of subassemblies and partly by extension of boundaries, should it be really necessary to cover these rather remote eventualities.

K. H. KÜSTERS: We do not feel that the effects of fission products are not very great in large reactors. On the contrary, for large reactors with a high burnup you may have a change in criticality of 4% or 5%. This may be greater for steam cooled reactors. But this is only the reactivity change due to the change from cold to the high burnup. In a steady or quasi-stationary state you do not have such a large spread in reactivity. The uncertainty in the fission product constants has led us to make a complete reinvestigation of fission product cross-sections, and we are just finishing these investigations now.

H. H. HUMMEL: It is not true that in the Okrent comparison higher isotopes were neglected because Pu^{240} was present in some of the mixtures. The uncertainty in the Pu^{240} capture cross-sections certainly played some role in some of the discrepancies, although it turned out to be somewhat less than the Pu^{239} fission cross-section and U^{238} capture cross-section. On the other hand, there seemed to be a possibility that there was some feedback between cross-sections assumed by different organizations for Pu^{240} capture. So perhaps the whole story was not really told there.

As to what we can do—we have a difficult problem in trying to assess the effect of the higher plutonium isotopes, fission product capture, and temperature on sodium void effect—and critical mass, for that matter. We are trying to see what we can do about these things in critical assemblies. For the higher plutonium isotopes we are attempting to get mixtures of Pu^{240} and Pu^{241} which is quite important. Certainly in England they have already made measurements with higher Pu^{240} content material.

As to fission products, I do not know what we shall do about them. It is an unsolved problem for critical assemblies. With regard to temperature, we have talked about trying to operate heated critical assemblies, but nobody is very enthusiastic about it. You can get some information on the effect of temperature on the sodium void effect by making Doppler effect measurements with or without sodium, because there is a relationship there which is helpful in trying to deduce the temperature dependence of the sodium effect. One would like to make direct measurements of this in a heated critical assembly, but it would be expensive, and we are not seriously considering this.

D. C. G. SMITH: Perhaps I might make two supplementary comments. Firstly, our paper refers to the change in enrichment as a result of adding fission products, control rods, and changing the Pu^{240} content quite considerably. So we have dealt with that to some extent.

Secondly, estimation of the effect of fission products will be very difficult in a critical assembly. In getting these fission products you cannot get away from something like 10 curies per gramme activity from the fission products themselves. I doubt whether any experimenter will be very happy about doing a perturbation in his core with such a highly active sample.

K. H. KÜSTERS: There will be a measurement of simulated fission product in the core of SNEAK. There is to be an investigation to simulate the absorption cross-section of the fission products, simulating them by appropriate materials.

R. D. SMITH (UKAEA): I am not very keen on the simulated fission product measurements. It seems to me that if you know what the cross-sections are, then you can calculate them and you do not need to simulate them.

A. G. EDWARDS (UKAEA): I should like to ask Dr. Storrer about three-dimensional codes. My particular interest is the Dounreay fast reactor, which is a small three-dimensional system, and I can, therefore, watch with a measure of detachment the dialogues about Doppler coefficient and sodium coefficient since our Doppler coefficient is virtually zero and our sodium coefficient is firmly negative. But when it comes to plotting flux distributions through our core, the situation is very different. The core is truly three-dimensional. The accuracy required is tied up with predicting the results for irradiation experiments and is greater than that required for running the reactor as such.

Is there any hope of getting a few group three-dimensional codes in the foreseeable future? How does Dr. Storrer's Monte Carlo code stand when it comes to predicting not reactivity effects but rather reaction rate distributions?

F. STORRER: We originally developed our Monte Carlo code for criticality safety problems. But, as I said, it does everything. It treats the three-dimensional problem and it treats heterogeneity at the same time. What is very interesting is that the running time is absolutely independent of the number of groups or of any fine structure, or any complication in the geometry.

That is because if you take the cross-sections from a ten-group set or from a 1000-group set, the neutron history will not differ. You just follow the neutron as it goes along the reactor.

It is also independent of the fine structure. It will take exactly the same time to get 0·2% accuracy in δk for a homogeneous reactor or for a heterogeneous one, essentially, because, no matter what the reactor comparison is, you need just so many neutron histories to achieve a good degree of accuracy. If you have a small perturbation, a pin head, which you put in the reactor, you will not describe with this code the fine structure of the flux in this pin-head because it is not important. You will not have enough neutron histories there and that proves that it is not important for reactivity.

With regard to reaction rates, I think that the code can describe them quite accurately, and much better than any existing method, close to the interface. I know of no method that can adequately treat the interface, except maybe one being developed by the Russians, the sub-group method, of which we heard in Geneva. Apart from that, I think that the Monte Carlo code is the only one.

I was also asked whether the fine structure of the flux is given accurately by the Monte Carlo method. The answer is not if you have many identical cells next to each other. The Monte Carlo code will give you a different fine structure in each cell because you do not have enough neutron histories per cell to have good statistics per case. But if you take an average of the fine structure over the 1000 cells, for instance, you will get a good fine structure within a cell, because you get better statistics.

So I think that the Monte Carlo method is practically identical to an experiment. We calculate well what is important and do not calculate what is not important.

R. D. SMITH: It seems to me that if you are interested in irradiating pin heads it is the reaction rate in the pin head that you are interested in, and, therefore, you must have some modification to the Monte Carlo code which will ensure that you get sufficient events in the pin head and its vicinity. It is this sort of code that we are looking for to deal with the problem of determining the environment of an irradiation experiment.

D. C. G. SMITH: As a back-up to Dr. Storrer's code, perhaps I might mention that at Winfrith there is under development a flux synthesis code of the sort that I think was developed at the Bettis Laboratory, by means of which we hope we shall be able to take a number of radial planes of the reactor and by some synthesis process with a set of trial functions, obtain reaction rates in three dimensions.

F. M. HECK (Westinghouse, U.S.A.) I believe that there is a few group three-dimensional code developed at Bettis. I do not know that it has been released. I believe that it also requires a Stretch computer to run, which may not make it very helpful to anyone else.

THE CHAIRMAN: Let us go back to the more practical problems. I should like to hear opinions about the necessity to check large fast reactors with the help of critical assemblies, bearing in mind that with the increase in size of assembly we shall get a greater difference between the model and the reactor owing to heterogeneity. It seems that the error when we transfer from the reactor to the model will not be less than the error of calculation.

H. H. HUMMEL: That is a matter which is of concern to us. We are constructing a large plutonium fuelled critical assembly ZPPR, but it will have mostly low Pu^{240} fuel. It will be operating at room temperature, and there will not be any fission products in it, so that there will be a long extrapolation. My feeling, which is not necessarily shared by any of my colleagues, is that for more of these higher isotope effects or effects of this type we shall have to rely on zoned experiments anyway to try to get the central worth effects for higher plutonium isotopes, in which case there may not be too much sense in constructing a complete critical assembly. The zoned assemblies certainly will play a very useful role. Exactly how much good we get out of building a 4000–5000 litre critical assembly which does not mockup the reactor too faithfully troubles me somewhat, too. I think that we have to do the sort of thing that Dr. Storrer suggests—get a fundamental understanding of what is going on, and after a certain point we just have to rely on the calculations.

D. C. G. SMITH: We have heard several papers describe reactor kinetics' measurements which have been made with perturbations other than the sinusoidal reactivity oscillator, even when the latter was the scheduled device. This may be because the reactor operators have not been convinced of the usefulness of the oscillator and have consequently given it a comparatively low priority in maintenance and installation programmes. In some ways I share their scepticism. Yesterday we saw the kinetic characteristics of the Fermi reactor and it had a very large bump, reminiscent of EBR-I, Mark ii. It is not easy to distinguish between this Fermi response, believed to be harmless, and the EBR-I response, known to be hazardous. It is small wonder that the operators are unconvinced of the worth of the oscillator. At AEEW, we have been examining the possibility of presenting the kinetic characteristic of the reactor in the time domain, and preliminary results suggest that this may provide a more easily interpreted picture of the state of the reactor plant, with a simpler criterion for safe operation.

N. V. KRASNOYAROV: As I understand it, we are speaking about the possibility of presenting the kinetic properties of the reactor in different ways. From the point of view of one group of characteristics the reactor may look unstable, but it may behave in a stable way.

My view is that representation by a Nyquist diagram, by amplitude and phase dependent on frequency, or by dependence of the feedback reactivity after a step in power, are all equal methods. If by one method we see that the reactor is stable, this also means that the reactor should be stable when we check it by other methods. It may be that on the basis of one characteristic we consider that the reactor is unstable, but in practical terms it is stable. This means that the characteristics do not correspond to the behaviour of the reactor. I would suggest that it is better to pay more attention to the methods of measuring characteristics.